

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Case Study

**COURSE NAME: Cloud Computing and
Big Data Analytics**

COURSE CODE: CSA1511

COURSE OUTCOME COVERED

CO3: Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)

Question Count: 3

Total Marks: 30

Code of Conduct

I **Nivasini VM 192511356** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Case Study

1. Case Study 1: Smart Retail Inventory Management System

A retail chain implements a cloud-based inventory system connected to store devices.Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewall and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

Case Study 1: Smart Retail Inventory Management System

1. Introduction

A Smart Retail Inventory Management System is a cloud-based solution used by retail stores to manage products, sales, stock levels, and customer transactions efficiently. It connects POS systems, store devices, dashboards, and centralized cloud services through APIs.

2. Cloud Architecture

Component	Function
Clients	POS systems and web dashboards used by employees
API Gateway	Handles and controls communication between clients and cloud services

Cloud Compute	Processes inventory and transaction-related operations
Cloud Storage	Stores product information, transaction logs, and analytics datasets
Platform Services	Provides analytics and demand-forecasting services
VPN	Provides secure communication between stores and cloud infrastructure
Firewall	Protects cloud resources from unauthorized network access
Web Application	Provides an interface for inventory monitoring and management

3. Working of the System

1. A customer purchases a product through the **POS system**.
2. The POS system sends transaction information to the API Gateway.
3. The cloud service processes the transaction and updates the inventory.
4. Product and transaction information is stored in cloud storage.
5. Platform-based analytics services analyze historical sales data.
6. Demand forecasting predicts future product requirements.
7. Store managers view inventory levels, sales reports, and predictions through a web dashboard.

8. VPNs, firewalls, and API gateways provide secure access to the system.

4. Cloud Service Models

- **IaaS (Infrastructure as a Service):** Provides virtual machines, networking, storage, and other infrastructure resources.
- **PaaS (Platform as a Service):** Provides analytics, databases, application development platforms, and demand-forecasting services.
- **SaaS (Software as a Service):** Provides the inventory management web application and dashboards to retail employees.

5. Advantages

- Real-time inventory monitoring
- Centralized data management
- Reduced manual work
- Improved demand forecasting
- Easy access from multiple stores
- Scalable cloud infrastructure
- Better security through VPNs, firewalls, and API gateways
- Reduced chances of stock-outs and overstocking

6. Conclusion

The Smart Retail Inventory Management System demonstrates how IaaS, PaaS, and SaaS can work together in a cloud environment. POS systems and dashboards communicate with centralized cloud services through APIs, while cloud storage, analytics, and forecasting services support efficient inventory management. Security mechanisms such as VPNs, firewalls, and API gateways ensure safe and reliable operations.

2. Case Study 2: Cloud-Based Document Collaboration System

An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices. Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

Case Study 2: Cloud-Based Document Collaboration System

1. Introduction

A Cloud-Based Document Collaboration System is an online platform that allows multiple users to create, edit, share, and manage documents through web browsers. It provides real-time collaboration, cloud storage, synchronization, and secure access from different devices and locations.

2. Cloud Architecture

Component	Function
Web Browsers	Allow users to access and edit documents
Web Application	Provides document creation, editing, sharing, and collaboration features

Web APIs	Synchronize documents and user activities across devices
Cloud Storage	Stores documents and related data in distributed storage
Version Control	Maintains previous versions and tracks document changes
Network Infrastructure	Provides low-latency and reliable global connectivity
Identity Management	Authenticates users and manages their identities
Access Permissions	Controls who can view, edit, or share documents
Encryption	Protects documents and communication from unauthorized access

3. Working of the System

1. Users log in to the platform through a web browser.
2. The identity management system authenticates the user.
3. The user creates or opens a document stored in distributed cloud storage.
4. Multiple users can edit the document simultaneously.
5. Changes are communicated through web APIs and synchronized across connected devices.
6. The system automatically maintains different document versions.

7. Users can share documents by assigning appropriate view or edit permissions.
8. Encryption protects data while it is being transmitted and stored.
9. Global network infrastructure provides low latency and high availability.

4. Cloud Service Models

- **IaaS (Infrastructure as a Service):** Provides servers, networking, virtual machines, and storage infrastructure.
- **PaaS (Platform as a Service):** Provides APIs, databases, synchronization services, and development platforms.
- **SaaS (Software as a Service):** The complete document collaboration application is provided to users through a web browser.

5. Advantages

- Real-time collaboration between multiple users
- Access from anywhere and from multiple devices
- Automatic document synchronization
- Version history and recovery
- Centralized cloud storage
- Easy document sharing
- High availability and scalability
- Secure access through authentication and permissions
- Reduced requirement for local storage

6. Conclusion

The Cloud-Based Document Collaboration System demonstrates how cloud computing enables real-time document editing and sharing. Distributed cloud storage provides reliable data storage, while APIs synchronize changes across devices. Encryption, identity management, and access permissions ensure security, making the system suitable for modern enterprise collaboration.

3. Case Study 3: Online Food Ordering Platform

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a clou platform. Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study 3: Online Food Ordering Platform

1. Introduction

An Online Food Ordering Platform is a cloud-based application that allows customers to browse food menus, place orders, make payments, and track their orders using web browsers or mobile applications. The platform uses RESTful APIs, cloud storage, virtual machines, and Kubernetes to provide a scalable and reliable service.

2. Cloud Architecture

Component	Function
Web Browser & Mobile Clients	Allow customers to browse menus and place orders
Frontend Application	Provides the user interface for customers and restaurants

RESTful Web APIs	Enable communication between frontend clients and backend services
Backend Services	Process orders, customer information, menus, and payments
Cloud Storage	Stores menu images, invoices, and customer data
Virtual Machines	Provide computing resources for backend applications
Managed Kubernetes	Manages and scales application containers
Authentication Tokens	Verify authenticated users
HTTPS	Encrypts communication between clients and servers
Role-Based Access Control (RBAC)	Controls access based on user roles

3. Working of the System

1. Customers access the platform through a web browser or mobile application.
2. The frontend displays restaurants, menus, prices, and food images.
3. Customers select food items and place an order.

4. The frontend sends the order information to backend services through RESTful APIs.
5. Backend services process the order and update the order status.
6. Cloud storage securely stores menu images, invoices, and customer information.
7. Virtual machines provide the required computing infrastructure.
8. Managed Kubernetes automatically manages application containers and provides scalability when the number of users increases.
9. Authentication tokens verify users, while HTTPS protects data during transmission.
10. RBAC ensures that customers, restaurant staff, and administrators can access only the functions permitted for their roles.

4. Cloud Service Models

- **IaaS (Infrastructure as a Service):** Virtual machines, networking, and computing resources are provided through the cloud.
- **PaaS (Platform as a Service):** Managed Kubernetes and other cloud platform services support application deployment, scaling, and management.
- **SaaS (Software as a Service):** The complete online food ordering application is delivered to customers through web and mobile clients.

5. Advantages

- Easy online food ordering
- Accessible through web and mobile devices
- Real-time order processing
- Scalable infrastructure
- Reliable application availability
- Secure storage of customer information
- Fast communication using RESTful APIs
- Automatic scaling using Kubernetes
- Role-based access improves security

6. Conclusion

The Online Food Ordering Platform demonstrates the integration of clients, RESTful APIs, cloud storage, virtual machines, and managed Kubernetes in a cloud environment. Security mechanisms such as authentication tokens, HTTPS, and RBAC protect users and data. Cloud-based infrastructure provides the scalability and reliability required to handle real-time food ordering operations efficiently.